# CS-117: Module Overview

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September 2026

# Lecturer

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- Office: Computational Foundry 310
- Office hours
  - Wednesdays, 1pm-2pm
  - Thursdays, 1pm-2pm

# Textbook

- Faron Moller, Georg Struth, Modelling Computing Systems, Springer, 2013.
  - 34 physical copies available from Bay Campus library.
  - $\infty$ electronic copies available from the university library website.

# Module Contents

- In this module you will be introduced to some of the formal mathematical methods that are essential for modelling, specifying, analysing, verifying, and validating computing systems.
  - These are key skills, especially in the "Age of AI".
- Further, this module will help you develop your general mathematical, analytical, and problem solving abilities.
  - This is essential in all areas of computing and beyond.

# Syllabus

Tentative schedule:

- Propositional Logic (Week 1 + 2)
- Sets, Functions, and Relations (Week 3 + 4)
- Predicate Logic (Week 5)
- Inductive Data Types (Week 6)
- Proof by Induction (Week 7)
- Games, Transition Systems, and Processes (Weeks 8 - 10)

# Module Components

- Lecture videos
- Lectures
- Problem sheets
- Labs
- Quizzes

# Lecture Videos

- Each week before the lectures, I will release one or two lecture videos (created by Prof Faron Moller).
- The lecture videos present the materials that will be covered that week.
- You should watch the lecture videos **before** the Wednesday lectures.

# Lectures

- I will discuss relevant aspects of the last week's assessment.
- I will go over the material in the lecture videos and answer your questions.
- We will solve exercises related to the material together.
- There will usually be small in-class quizzes to facilitate this.

# Problem Sheets

- Every week I will release one or two problem sheets.
- The problem sheets will contain a large number of problems.
  - You will not be expected to solve all problems on the problem sheets.
  - I will indicate a selection of problems you are expected to solve.
- This is a problem-solving subject!
- Solving the problems on the problem sheets is the most important part of the module!
- The main purpose of the lectures is to provide you with the necessary background you need to tackle the problems.

# Problem Sheets

- You are **not** formally assessed on the problem sheets...
- ...but solving them will be essential for the weekly quizzes and the final exam.
- I will release solutions to all problems on the problem sheets.
- Only check the solutions once you have found a solution yourself!
- Similarly, do not use AI or Google to "help" you with the problems.
- If you are "stuck" on a problem, discuss it with your coursemates or teaching staff first, e.g. during lab sessions or office hours.

# Weekly Labs

- We have labs every week on Thursdays, 2pm - 4pm.
- You can (but do not have to) use the labs to work on the problem sheets.
- You *should* use the labs to discuss your solutions to the problems with teaching staff (the lecturer or TAs).
- You can (but do not have to) complete the weekly quiz during the lab hours using one of the lab machines.

# Quizzes

- There will be weekly quizzes on Canvas.
- Your quiz scores contribute 15% to your final grade.
  - We will have ~10 quizzes, so each individual quiz will contribute ~1.5% to your final grade.
- The quiz will usually be released on or before Monday each week.
- You have to start the quiz before Friday, 4.30pm.
- Once started, you have 30 minutes to complete the quiz.
  - You may have reasonable adjustments that allow you to complete the quiz in 50 minutes.
  - You will know whether you have adjustments. When in doubt, please shoot me a quick email to confirm.
- Modulo the above constraints, you can attempt the quiz anywhere and at any time.
- You can (but do not have to) use one of the lab computers during the PC lab to complete the quiz.

# Quizzes

- The quizzes are designed to test your knowledge and understanding.
- You should attempt them **after** watching the lecture videos, attending the lectures, and solving the weekly problem set.
- You must complete the quizzes by yourself.
- *i.e.*, you are not allowed to elicit help from
  - Coursemates
  - Google
  - LLMs
  - etc.

# Workflow

- Before the Wednesday lecture:
  - Watch the lecture videos.
  - Have a look at the slides for the week.
  - Have a look at the problem sheet(s) for the week.
  - Ideally, attempt some of the problems on the problem sheet.
  - Prepare questions.
- During the lectures:
  - Ask questions.
  - Participate in the in-class exercises.
- After the Wednesday lectures:
  - Start working on the problem sheets.
  - Discuss your solutions with teaching staff (TAs/lecturer) during lab hours and/or office hours.
  - Complete the weekly quiz **before Friday, 5pm.**

# Assessment

Assessment components:

- Final Exam (January): 65%
- Weekly Quizzes ("Laboratory Work"): 15%
- Coursework (November): 20%
  - Released on or before Friday, 13th November, 2pm.
  - Due Friday, 27th November, 2pm.
  - Feedback released on or before Friday, 11 December, 2pm.